

2D SCANNING LIDAR · LR-1BS SERIES

30-METRE OLEI LR-1BS5+ 270° LIDAR

The longest reach in the LR-1BS family. Target enhancement technology pushes a 10% reflectivity target out to 15 metres and an 80% target to 30 — in the same 450-gram, 60 × 60 × 85 mm body as the rest of the series.



■ WHY THIS SENSOR

- **0.2–30 m at 80% reflectivity** — the longest range in the LR-1BS family, and 30 m on reflectors too
- **Target enhancement technology** — a 10% reflectivity target reaches 15 m — nearly double the standard LR-1BS5
- **Built for asphalt and plant rooms** — the low-reflectivity surfaces that defeat a standard scanner
- **User-settable weak-signal threshold** — tune sensitivity to the site without changing the mount
- **Built-in multi-level filtering** — configurable tailing suppression at the sensor, so the cloud arrives clean
- **< ±20 mm absolute accuracy on common targets** — < ±40 mm on reflectors, ≤ 20 mm repeat
- **40K points per second, 0.225° fixed resolution** — azimuth, distance and reflection intensity over 100 Mbps Ethernet
- **450 g, IP65, 5 W at 12–24 V DC** — M8 power and LAN on 2 m leads, OLAMViewer and a ROS driver

0.2–30 m

MEASUREMENT RANGE

15 m

AT 10% REFLECTIVITY

< ±20 mm

ABSOLUTE ACCURACY

40K

POINTS / SECOND

■ APPLICATION SCENARIOS



SLAM NAVIGATION

Point-cloud mapping and localisation for mobile platforms



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles



DOCKING & TRACKING

Reflector-based docking and pallet or object tracking

Parameter	Technical data
Switching outputs	1
Ethernet connection	100 Mbps Ethernet
Connection	5-pin M8×1 female, 2 m (power) 4-pin M8×1 female, 2 m LAN
Operating voltage	12–24 V DC
Power consumption @ 25 °C	5 W
Mechanical dimensions	60 × 60 × 85 mm
Weight	450 g
Degree of protection	IP65 or optional
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Air humidity	< 95% RH
Optical indicators	RGB × 4
Ambient light limit	80 000 lux
Mounting	Bottom / back
Basic software	OLAMViewer · ROS driver

Technical drawing of the LR-1BS ultrasonic flowmeter, showing three views: front, top, and side.

Front View Dimensions:

- Overall width: 68.8±0.1 mm
- Overall height: 85 mm
- Top section height: 24.5 mm
- Middle section height: 46.0 mm
- Bottom section height: 14.0 mm
- Base thickness: 8.5 mm
- Base width: 35.0 mm
- Base gap: 0.4 mm

Top View Dimensions:

- Overall width: 76.0 mm
- Overall height: 60.0 mm
- Sensor array width: 51.0±0.1 mm
- Sensor array height: 60.0 mm
- Sensor array height: 33.0±0.1 mm
- Base thickness: 8.0 mm
- Base width: 35.0 mm
- Base gap: 0.4 mm
- Hole dimension: 3.0±0.05 mm
- Hole depth: 1.5 mm

Side View Dimensions:

- Overall height: 50.5 mm
- Base width: 17 mm
- Base width: 8.9 mm
- Base width: 4.5 mm
- Base width: 10 mm
- Base width: 2.9 mm

Other Details:

- Label: LR-1BS
- Serial Number: 84-40-07-60-00-00
- Manufacturer: M1P2019041000001
- Angle: 315°

UNIT: MM · DRAWINGS NOT TO SCALE